Anuj Gupta

Autumn Mid-Semester Examination 2019

EC2L002-Signals and Systems

Reg. Number

Duration: 2 Hour

Maximum Marks: 60

* All parts of answer together. Assume parameters if it is required with proper justification.
* All steps must be presented for the question.
* **Answers All Questions**

PART-I (3 × 6 =18 Marks)

1. Consider a casual LTI system implemented as the RLC circuit shown in Fig. 1. In this circuit, $x(t)$ is the input voltage. The voltage $y(t)$ across the capacitor is the output response of the system. (a) Find differential equation relating $x(t)$ and $y(t)$. (b) Determine the frequency response of this system by considering the output of the system to inputs of the form $x(t) = e^{j\Omega t}$. (c) Find the output $y(t)$ if $x(t) = \sin(t)$.

$R = 1\,\Omega$
$L = 1\,H$
$C = 1\,F$

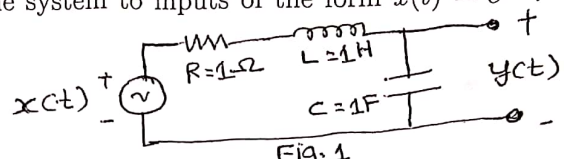

Fig. 1

2. (a) Compute the autocorrelation of the signal $x[n] = a^n u[n]$, where $0 < a < 1$. (b) Sketch the normalized autocorrelation sequence for $a = 0.5$. (c) Show that it is an even function.

3. Sketch the output of a LTI system with input $x[n] = 2^n u[-n]$ and unit impulse response $h[n] = u[n]$.

PART-II (10 × 2 = 20 Marks)

4. The continuous-time LTI system performs a pure time-shift operation which is described as $y(t) = x(t - t_o)$. Find the impulse responses of the system and its inverse system.

5. Prove the Parseval's theorem by using the sine waveform $x(t) = 4\sin(50\pi t)$. What is the average power of the $x(t)$?

6. Find the impulse response of a relaxed DT system which is described by $y[n] = x[n] - 0.5y[n-1]$.

7. Determine whether the signal $x(t) = \cos(\frac{2}{3}t + 30^o) + \sin(\frac{4}{5}t + 45^o)$ is periodic. If it is periodic, find the fundamental frequency and the period. What harmonics are present in $x(t)$.

8. Prove that response $y(t)$ of an LTIC system to a periodic input $x(t)$ is also periodic with the same fundamental period as $x(t)$ with variations in amplitude and phase.

9. Describe **Gibbs phenomenon** with a suitable example.

10. The convolution integral has several interesting properties that can be used to simplify the analysis of LTIC systems. State **six** properties of convolution.

11. Write **Dirichlet conditions** for the existence of the CTFS representation. Also provide an example for not satisfying each condition.

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12. Let the transmitted RF pulse be given by $x(t) = \begin{cases} \sin(\Omega_c t) & 0 \ll t \ll T_o \\ 0 & Otherwise \end{cases}$, where $T_o = 4T_c$ Sketch the transmitted signal. Suppose we transmit an impulse from the RADAR to obtain the impulse response of a round-trip propagation to the target that results in the impulse response $h(t) = \delta(t - D)$, where a represents the attenuation factor and D represents the round-trip time delay. Obtain the received signal and sketch the signal.

13. Describe eigenfunctions and eigenvalues of LTI systems with the suitable example(s).

PART-III (22 × 1 = 22 Marks)

(Please write only answers for the questions)

14. The relationship between the input and output of an accumulator is described as_____, which is an example of_____(FIR/IIR) system.
15. The periodic impulse train with a period T_o can be expressed as_____.
16. The difference equation of a non-causal M-point moving averaging system is described as_____.
17. The examples for singularity function are_____and_____.
18. The moving averaging filter is often used to filter out_____(high-frequency /low-frequency) noise from a signal that can be viewed as_____(low-pass /high-pass) filter.
19. The first-order backward difference is expressed as_____,which is often used to emphasize_____ portions in the signal.
20. The sampling property of the delta function is defined as_____.
21. The $\int_{-\infty}^{\infty} \delta(\frac{t}{3} - 1)\cos(10t)\,dt$ is equal to_____.
22. The CT sinc function is defined as_____,which is an_____(even/odd) function.
23. In digital signal processing, decimation is the process of_____the sampling rate of a signal that is the_____(irreversible/reversible) process.
24. Random signals are generally characterized by statistical measures such as_____,_____,_____, and _____.
25. In the analysis or design of a system, **six** basic categories of systems are: _____.
26. In class, we have studied about stationary and non-stationary signals. Based upon our discussion, _____and_____biosignals are examples for cyclostationary signals in normal recording conditions.
27. The practical examples for time-variant system are_____and_____with specific observation interval.
28. Determine whether the CT integrator system is stable.
29. The audible frequency range is_____and the audio signal is recorded by using_____sensor.
30. In class, we discussed many basic (elementary) signals. _____and_____are orthogonal signals.
31. Write two advantages and disadvantages of digital signal processing.
32. In class, we have studied that Smartphone has many sub-systems. What are wireless communication modules embedded on the Smart-phone?
33. In class, we have studied many properties of CTFS. Write **eight** properties of CTFS.
34. The real-world signals having multiple frequencies are_____,_____,_____and_____
35. The four types of frequency-selective filters are_____,_____,_____and_____.

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